Discrimination of Depression Levels Using Machine Learning Methods on EEG Signals

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***Abstract*—Depression is a mental disorder which has direct effects on electroencephalography (EEG) of patients, that made EEG analysis a beneficial way for a depression diagnosis. A precise system which can diagnose the depression levels based on the EEG signal would be useful support. This paper presents a machine learning approach to discriminate the depressed subjects to four different levels of depression, according to the Beck depression inventory (BDI-II) scores, besides the separability of different levels is investigated. In this way, we also proposed a fuzzy function based on neural network (FFNN) classifier. Our dataset contains EEG signals recorded from 60 depressed subjects with different levels of depression, under resting state, and EEG analysis was done using nonlinear features including fuzzy entropy (FuzzyEn), Katz fractal dimension (KFD) and fuzzy fractal dimension (FFD). The results indicate that KFD has a better capability in the prediction of the depression level. The proposed fuzzy classifier has demonstrated significant supremacy compared to support vector machine (SVM) in almost all experiments.**

***Keywords*—Depression, Fuzzy Function, Nonlinear Systems, EEG, Fuzzy Entropy, Fractal Dimensions.**

I. Introduction

Depression is a growing mental disorder all around the world which has direct effects on brain performance of patients. Psychologists and psychiatrists use interviews to diagnose depression. Detection is performed using a diagnostic standard called Diagnostic and Statistical Manual of Mental Disorders (DSM) [1]. Various self-rating questionnaires have been used to measure depression level; Beck Depression Inventory-‖ (BDI-‖) is one of the most common tools used for this purpose [2].

Recently, diagnosis of mental diseases using a biomarker based on EEG has spread out especially in case of depression disorder [3]. In these studies, the type of features (linear or nonlinear) extracted from the EEG signal is investigated. Since the EEG signal is a nonlinear, complex, and non-stationary signal, the non-linear dynamics methods are widely used for analysis of chaotic behaviors of EEG [4-6]. Entropy is a phenomenon which in some views has a similar basis to complexity. Faust et al. reported a classification accuracy of 99.5%, using various entropy features extracted from EEG signals of normal and depressed patients [7]. Entropy has been employed in EEG signal analysis to investigate nonlinear behavior of the brain, fuzzy entropy (FuzzyEn) is one of the entropy calculation methods that is modified by an approximate entropy algorithm that has less sensitivity to noise [8]. Xiang et al. [9], applied FuzzyEn and Sample entropy to identify the epileptic seizure and concluded that FuzzyEn is a better index.

Fractal dimension (FD) is another measure that represents the EEG signal's complexity [10]. FD appeared as a powerful method in the EEG analysis, e.g., to identify different sleep stages [11]. Raghavendra et al. demonstrated that FD is a useful parameter for the characterization of schizophrenia diseases [12]. There are several algorithms have been proposed for estimation of the FD, e.g., Higuchi, Katz and fuzzy FD [13-15]. In literature, Katz fractal dimension (KFD) employed in EEG analysis applications in bipolar mood disorder and attention deficit hyperactivity disorder [16]. Pedrycz et al. introduced a method to estimate FD using fuzzy sets and made a description of self-similarity of time series [15].

Most of the studies discriminated normal and depressed patients [17, 18]. Hossienifard et al. [18] displayed the effectiveness of nonlinear analysis of EEG signal for depression with extracted four nonlinear features including detrended fluctuation analysis (DFA), Lyapunov exponent, correlation dimension and Higuchi fractal. They obtained 90% accuracy with the logistic regression classifier and the combination of features. Acharya et al. [17] presented a computer-aided nonlinear technique to diagnose depression using nonlinear features including: Hurst's Exponent (H), FD, Largest Lyapunov Exponent (LLE), DFA, Recurrence Quantification Analysis (RQA), and Sample Entropy extracted from EEG which were applied to SVM classifier. The best result was 98%.

In recent studies distinguishing between depressed people and healthy controls was investigated. The depression level diagnosis can help the psychiatrist start the treatment process appropriate for the individual. Therefore, in this study, we proposed a method to diagnose depression levels (which contains four levels including minimal, mild, moderate and severe, according to BDI-II). We employed nonlinear features including FuzzyEn and fractal dimensions (KFD and FFD). We applied a fuzzy function based on neural network (FFNN) classifier and support vector machine (SVM).

II. Materials and Methods

A. Data acquisition and preprocessing

The dataset contains EEG signals recorded from 60 participant (30 males, 30 females; 32.4 ± 10.5 years old) with different levels of depression disorder (the number of subjects for each level: minimal=17, mild=10, moderate=10,

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severe= 23). The 19-channel EEG recorded at Atieh Psychiatry Centre, Tehran, Iran with a sampling rate of 256 Hz for 4 minutes in the rest state with closed eyes. Depression diagnosis was done by a psychiatrist based on DSM-IV and depression level was determined using the BDI-II score.

A sequence of procedures has been applied as preprocessing to prepare processed data. A 2nd order high-pass zero-phase Butterworth filter with 0.5 Hz cutoff frequency and a similar low-pass filter with a cutoff in 45 Hz were employed due to the eliminating low and high-frequency noises. Since the low-pass filter was not sharp enough, a notch filter was used to remove the 50 Hz frequency effects on EEG signals. Independent component analysis (ICA) with the manual method was used to remove ECG, EOG and EMG artifacts. In this method, first, EEG signal was decomposed into some independent components (equal to the number of channels), and then the components that are most similar to the artifacts are eliminated. In the next step, the remaining components are returned to the sensor level. All signals were segmented to 6-second windows. The Features calculated from each window were averaged over all windows. To avoid high dimensionality problems, PCA with Gaussian kernel was used to reduce the input dimension to 15. All of the analysis and processing were done by MATLAB.

B. Feature extraction

The brain is a nonlinear and complex system, so nonlinear methods are proper tools to describe brain dynamics and behaviors. In this way, we employed three nonlinear features explained as follows:

1) Fuzzy entropy

Entropy is a measure to investigate the dynamic complexity of time series. Fuzzy entropy is an improved method which is employed to understand complex and nonlinear system behaviors [19]. The calculation of FuzzyEn is described as follows:

For an N sample time series of EEG signal, $\{u(i):1 \le i \le N\}$, given m, and a set of an m-dimensional vector, $\{X_i^m, i=1,...,N-m+1\}$ $(m \le N-2)$ can be calculated:

$$X_i^m = \{u(i), u(i+1), ..., u(i+m-1)\} - u_0(i) \tag{1}$$

Where $u_0(i)$ represents the average value, which is defined as:

$$u_0(i) = \frac{1}{m}\sum_{j=0}^{m-1} u(i+j) \tag{2}$$

Then, define the distance d_{ij}^m between two vectors, X_i^m and X_j^m, as the maximum absolute difference of corresponding scalar components of two vectors:

$$d_{ij}^m = d[X_i^m, X_j^m] = \max_{k\in(0,m-1)}\{|(u(i+k)-u_0(i))- (u(j+k)-u_0(j))|\} \tag{3}$$

, $(i, j = 1 \sim N-m, j \ne i)$

Calculate the similarity degree D_{ij}^m of X_i^m and X_j^m using fuzzification of d_{ij}^m with fuzzy membership function $\mu(d_{ij}^m, n, r)$ which defined as:

$$D_{ij}^m = \mu(d_{ij}^m, n, r) = \exp(\frac{-(d_{ij}^m)^n}{r}) \tag{4}$$

Where r and n are the width and the gradient of the exponential function, respectively.

Define the function $\varphi^m(n,r)$ as:

$$\varphi^m(n,r) = \frac{1}{N-m}\sum_{i=1}^{N-m}\left[\frac{1}{N-m-1}\sum_{j=1, j\ne i}^{N-m} D_{ij}^m\right] \tag{5}$$

Finally, the fuzzy entropy can be calculated using the following formula:

$$FuzzyEn(m,n,r,N) = \ln\varphi^m(n,r) - \ln\varphi^{m+1}(n,r) \tag{6}$$

According to [9], m=2 and $r = 0.25 \times sd$ considered for FuzzyEn parameters, which sd is standard deviation of N-point time series.

2) Katz fractal dimension

Fractal dimension presents an index which demonstrates the complexity of signal [10]. Katz is an algorithm to calculate the fractal dimension based on variations of time series' amplitude. Katz index is calculated as follows:

$$D_k = \frac{\log_{10}(L)}{\log_{10}(d)} \tag{7}$$

Where L is the length of the time series, and d is the distance between the first point of sequence and the farthest point [14]. d , is described as:

$$d = \max\{\text{distance}(1, i)\} \tag{8}$$

Where i is a data point that maximizes distance from the first point of a sequence to this point.

3) Fuzzy fractal dimension

In the phase space of dynamical systems, the dimension is a concept which describes the complexity of a system about the independent variables needing to system description. Fractal dimensions demonstrate the self-similarity or structural complexity of a particular time series, and FFD employs fuzzy sets to construct fractal dimensions [15].

In the conventional method, hyperspheres were constructed around each point of signal and then the total number of points existing inside a sphere with a radius ε accounts for $N(\varepsilon)$. But in FFD, we use a fuzzy set for scaling in the computations.

$$N(\varepsilon) = \frac{1}{N(N-1)}\sum_{i=1}^{N}\sum_{\substack{j=1\\ j\ne i}}^{N}\Omega_{ij}(\varepsilon)A_{ij}(\varepsilon) \tag{9}$$

$$A_{ij}(\varepsilon) = \exp(-(x(i) - x(j))^2 / \varepsilon^2) \qquad (10)$$

Where A_{ij} , considered as a Gaussian membership function centered at $x(i)$. Now the FD calculated with fuzzy sets is defined as follows:

$$\log N(\varepsilon) = D \log \varepsilon + C \qquad (11)$$

It can be shown that the FFD is the slope of the regression line fitted to the experimental points, $(\varepsilon_j, N(\varepsilon_j)), j = 1,...,c$.

In this method, each data point inside the sphere contains a membership value that depends on its distance to the center. Therefore, greater A_{ij} value (membership value) for an individual $x(j)$, leads it to have a more significant effect in total sum $N(\varepsilon)$.

C. Proposed Model

We proposed a fuzzy function based on neural network (FFNN) model for data classification. The schematic of the model has been shown in Fig. 1, note that this schematic shows a four-class classifier and we also designed a binary version similarly, their applications discussed in section III.

First of all, the input data ($\mathbf{X}_{N*d}$ matrix, where N , is number of data points and, d is the dimension of input data) divided to c clusters using Gustafson-Kessel (GK) clustering algorithm [20]. The parameters of GK, c and f are the number of clusters and degree of fuzziness respectively, the optimum values determined with several experiments $c = 15, f = 2$. GK algorithm give a vector of membership values $\boldsymbol{\mu} = [\mu_1, \mu_2, ..., \mu_c]$ for each data point. Then, according to the Türkşen method [21], each of membership values is squared and $\boldsymbol{\mu}^2$ is placed beside the $\mathbf{X}$ to form $\mathbf{X}' = [\mathbf{X}, \boldsymbol{\mu}^2]_{N*(c+d)}$. Now we use $\mathbf{X}'$ as an input matrix for a one-hidden-layer neural network (NN) that is a nonlinear function $f : R^{(d+c)} \to R^c$, where $(d+c)$ is the dimension of the input matrix and c is the dimension of the output. The equations of NN layers are as follows:

$$\mathbf{U} = \varphi(\mathbf{X}' * \mathbf{W}_1) \qquad (12)$$

$$\mathbf{V} = \varphi(\mathbf{U} * \mathbf{W}_2) \qquad (13)$$

Where φ is a sigmoid function. $\mathbf{U}_{N*l}$ is the output of first layer which l was selected equal to 15 and $\mathbf{V}_{N*c}$ is the output of second layer. Both $\mathbf{V}$ and $\boldsymbol{\mu}$ have the same dimension and $\mathbf{Z}$ calculated as follows:

$$\mathbf{Z} = \mathbf{V}.\boldsymbol{\mu} \qquad (14)$$

Where (.) an element by element product. This layer performs as a feature selection algorithm for the next layer. If the data point has a weak cluster membership the multiplication of input data's membership values by the NN's output node decreases the effectiveness of the corresponding node of NN.

Finally, a *Softmax* function eq. (15) is applied to derive the probability values of data points of belonging to each class:

$$y(z)_j = \frac{e^{z_j}}{\sum_{k=1}^{K} e^{z_k}} \quad for \quad j = 1,...,K \qquad (15)$$

$$\mathbf{Y} = Softmax(\mathbf{Z} * \mathbf{W}_3) \qquad (16)$$

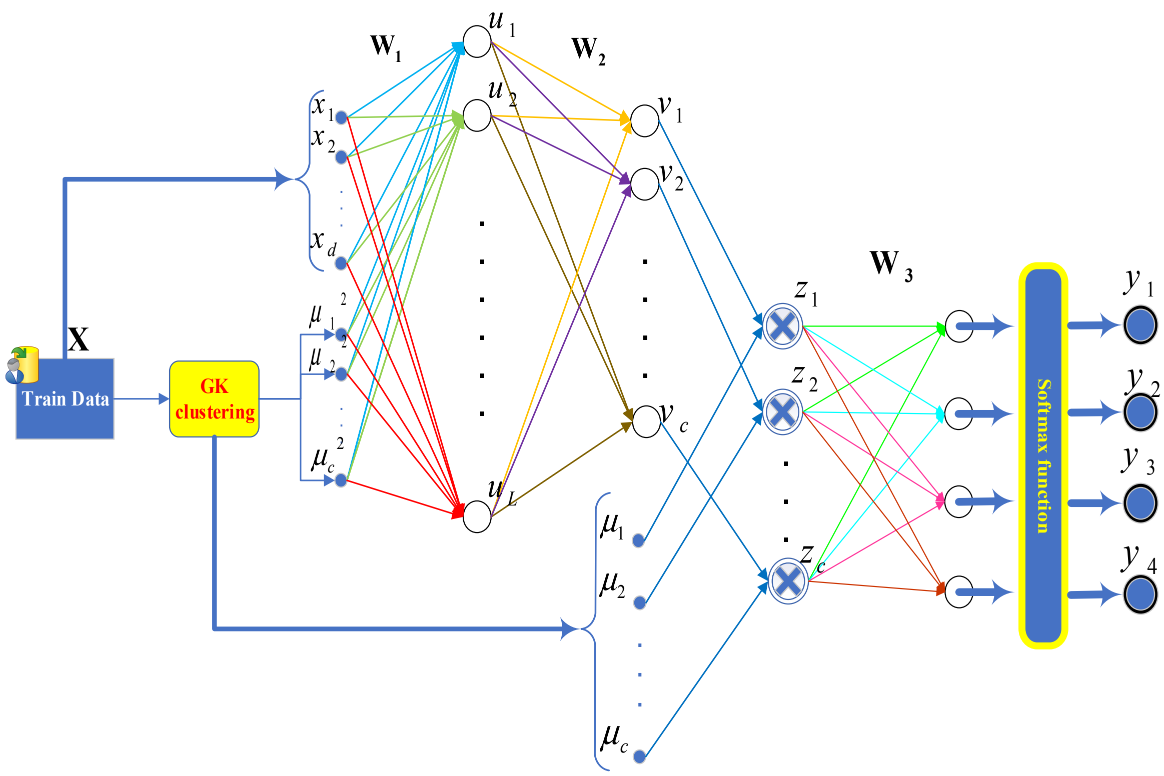
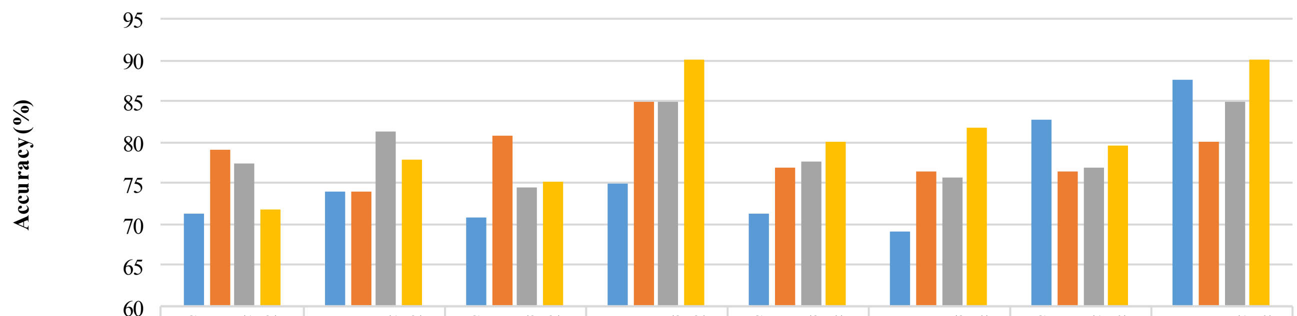


Fig. 1. A conceptual model of the fuzzy function based on neural network (FFNN) classification. GK refers to Gustafson-Kessel clustering method and μ_c is the membership of c th cluster.



	SVM (1-2)	FFNN (1-2)	SVM (2-3)	FFNN (2-3)	SVM (3-4)	FFNN (3-4)	SVM (1-4)	FFNN (1-4)
FuzzyEn	71.22	74.08	70.75	75	71.28	69.05	82.73	87.5
FuzzyFractal	78.97	74.07	80.85	85	76.96	76.5	76.52	80
KATZ	77.3	81.33	74.41	85	77.72	75.75	76.97	85
All Features	71.85	77.77	75.17	90	80.15	81.78	79.46	90

Fig. 2. Comparison of classifiers' accuracy for each nonlinear feature and combined nonlinear features in depression levels.

where $\mathbf{W}_3(c*k)$ is Task-driven part's weight and $\mathbf{Y}(N*k)$, which is the predicted labels of the model with k classes. To set model parameters, first we considered a random number for the weights between all the layers $(\mathrm{w}_1, \mathrm{w}_2, \mathrm{w}_3)$ and calculated the gradient for each parameter in the cost function, then parameters were optimized with the back-propagation algorithm:

$$\phi = \frac{1}{2}\left\|\hat{\mathbf{Y}} - \mathbf{Y}\right\|_2^2 \tag{17}$$

$$\frac{\partial \phi}{\partial \mathbf{W}_l} = -\left|\hat{\mathbf{Y}} - \mathbf{Y}\right|\left(\frac{\partial \mathbf{Y}}{\partial \mathbf{W}_l}\right) \tag{18}$$

$$\mathbf{W}_l^{new} = \mathbf{W}_l^{old} + \alpha\frac{\partial \phi}{\partial \mathbf{W}_l} \tag{19}$$

Where ϕ is a cost function and $\alpha < 1$ is the learning rate that in this study we set $\alpha = 0.04$.

III. Results and Discussion

In this study, three nonlinear features were extracted from EEG signals of 60 subjects with different levels of depression. Then, SVM and FFNN classifier were employed to discriminate depression levels. Leave-one-out cross-validation (LOOCV) technique is used to validate the model. In LOO, all subjects except the last one are exploited to train a model, and the remaining subject is validated based on the trained model.

We applied the binary classifier to some pairs of depression levels to investigate the separability of different levels. As shown in fig. 2, using FFNN classifier, for FuzzyEn the most separability reported from the classification accuracy of 1-4 pair (note: numbers refer to depression levels or classes, including 1: minimal, 2: mild, 3: moderate, 4: severe) is 87.5%. For FFD feature, the best accuracy was 85% which has been obtained from 2-3 pair by FFNN classifier. Also, FFD shows the best performance in classification of 1-4 pair in comparison with all features. The Katz fractal dimension resulted in the best accuracy of 85% for both 1-4 and 2-3 pairs. The accuracy of the classification of pairs of groups 1-4 and 2-3 with the combination of features has increased by 90%. Finally, we can say that proposed FFNN classifier demonstrated more significant performance than SVM in most classification procedures. Also Fig. 2 shows high average accuracies obtained when all features together were fed to classifiers as input.

On the other hand, we considered multi-class SVM and FFNN classifier as a system to identify the subjects' depression level directly, using the three nonlinear features. Fig. 3, shows that Katz FD had the best accuracy in comparison with two other features. Also, fuzzy classifier had better performance than SVM. Again, by combining the three features, the results of the two classifications were improved, with superiority for FFNN classifier.

In the classification of minimal-severe depression level,

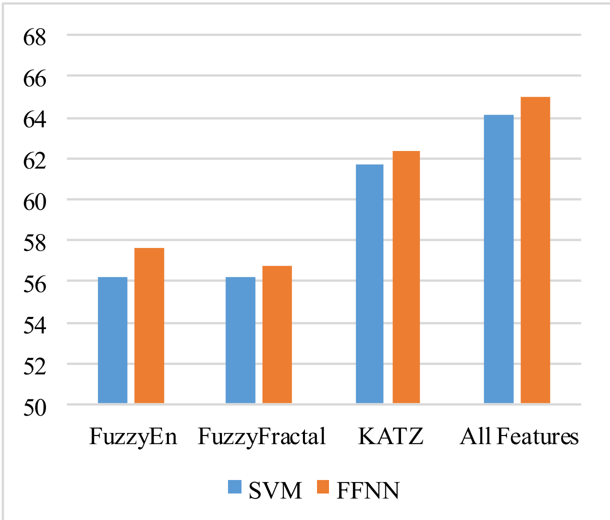


Fig. 3. The accuracy of four depression levels' classification using SVM and FFNN model for each nonlinear feature and all features together.

the highest accuracy for a single feature type procedure resulted when we used FuzzyEn with FFNN classifier. This accuracy was improved even more and reached 90% when all features together were applied to the model. The experiments revealed the ability of nonlinear features in discrimination of depression levels that an acceptable accuracy achieved from only a single nonlinear feature. The results showed a significant difference between the complexities of different depression levels.

IV. CONCLUSION

In this study, we introduced a machine learning approach to diagnosing the depression level using EEG signals of depressed subjects and investigated the separability of different levels of depression from each other. To analyze EEG signal some nonlinear features were extracted from signals after cleaning them from noise and artifacts. In the classification of depression levels, Katz fractal dimension showed the most significant performance. The accuracy increased when all features were employed together. According to experiments on an investigation of pairs of depression levels, minimal and severe levels are more discriminated in comparison to other depression level pairs.

We also proposed a fuzzy function based on a neural network model which showed a prominent advantage in comparison to SVM. The proposed model is a combination of the neural network with the fuzzy clustering and provides a nonlinear model that makes it possible to differentiate data into different classes.

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